

Dual-Fuel Reactor Network Technical Reference

Jet-A and LNG Combustor Modeling for Nozzle and Architecture Design

Beginner-Oriented Equations, Algorithms, Source Locations, and Limitations

fuelnozzle project

Draft, in progress

Abstract

This report explains the equations and software implementation of the reactor-network extension to the `fuelnozzle` package, for a reader with no prior background in combustion, chemical kinetics, spray physics, or Python. The extension takes the output of the Jet-A pressure-swirl and flashing-LNG nozzle models and carries it into the combustor, so that emissions, temperature, mixing, and autoignition risk can be estimated without a three-dimensional simulation. Its purpose is to support the design of a dual-fuel combustor that burns Jet-A during landing and take-off and LNG during cruise, with separate injector hardware for each fuel and a shared combustor. Every major equation is connected to the Python function that implements it, and physical law, numerical approximation, and empirical calibration are kept visibly separate throughout. This is documentation for a reduced-order screening and design-exploration tool, not a certification method.

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1 Status of This Document

This document is written alongside the implementation, section by section, as each module is completed. Sections marked *pending* are planned but not yet written, and their absence means the corresponding code does not exist yet either.

The companion documents are:

- `docs/CRN_PLAN.md`, the design intent, task list, and risk register;
- `docs/CRN_IMPLEMENTATION_LOG.md`, the build record, deviations from the plan, and verification evidence;
- `docs/technical_reference.tex`, the equivalent reference for the nozzle models that this extension consumes;
- `mech/README.md`, the kinetic mechanism registry and provenance.

2 How to Read This Report

Pending. Will follow the structure of the nozzle technical reference: separating what is a physical law, what is a numerical approximation, what is an empirical calibration, and what is simply unmodeled.

3 Basic Vocabulary for a New Reader

Pending. Reactors, residence time, equivalence ratio, ignition delay, emission index, and the difference between a reactor network and a computational fluid dynamics simulation.

4 Why a Reactor Network, and What It Cannot Do

Pending.

5 Kinetic Mechanisms

5.1 What a mechanism is

A *chemical kinetic mechanism* is a list of the molecules a fuel can turn into and the reactions that convert one into another, each with a rate. Burning a hydrocarbon is not one reaction but hundreds acting at once, so the mechanism is a file, not a formula. The files used here contain 47 to 71 chemical species and 247 to 538 reactions.

Two properties of a mechanism matter for this work. First, whether it contains *nitrogen chemistry*: without species such as NO and NO₂, the mechanism cannot produce oxides of nitrogen at all. Second, whether it contains *low-temperature chemistry*: the slow reactions that determine how long a fuel-air mixture can sit at moderate temperature before it ignites on its own.

5.2 Why Jet-A needs two mechanisms

Neither available Jet-A file has both properties, and this tool needs both.

File	Species	Nitrogen chemistry	Low-temperature chemistry
A2NOx_skeletal	71	yes	no
A2NTCfast_ske	47	no	yes
gri30 (LNG)	53	yes	n/a

It would be convenient to use the nitrogen-bearing file for everything. The following measurement, made during implementation, shows why that would be unsafe. The table gives the ignition delay predicted by each Jet-A mechanism at conditions representative of a premixing passage, expressed as the ratio between them.

Pressure	700 K	800 K	900 K	1000 K
20 atm	510×	71×	9.3×	2.0×
40 atm	490×	86×	13.5×	2.9×

Compressor discharge temperature at landing and take-off is roughly 700 to 900 K. In that range the high-temperature-only mechanism predicts an ignition delay one to two orders of magnitude longer than the mechanism that includes low-temperature chemistry.

Engineering caution. An ignition delay that is predicted too long makes a premixing passage look safe when it is not. This tool therefore uses `A2NOx_skeletal` for the reactor network and all NO_x output, and `A2NTCfast_ske` only for ignition delay. Each result records which mechanism produced it.

5.3 A distinction that is easy to get wrong

Some hydrocarbons show a *negative temperature coefficient*: a temperature band in which raising the temperature makes ignition take *longer*, contrary to intuition. Measured for `A2NTCfast_ske`, that band is clear at 5 atm between roughly 800 and 950 K, weak at 10 atm, and gone by 20 to 40 atm.

It is tempting to conclude that low-temperature chemistry stops mattering at combustor pressure. It does not. The 510× measurements above were made at 20 and 40 atm, where the non-monotonic shape has already disappeared. What vanishes at high pressure is the *shape* of the curve, not the *influence* of the low-temperature reactions on the absolute delay. These are two different statements.

5.4 Nitrogen pathway coverage differs between fuels

Nitric oxide forms by several competing routes. The two mechanisms in use do not cover the same set, and neither contains the other.

Route	A2NOx_skeletal (Jet-A)	gri30 (LNG)
Thermal (Zeldovich)	yes	yes
N ₂ O route	yes	yes
Prompt via NCN (modern)	yes	no
Prompt via HCN (older)	yes	yes
NNH route	no	yes

Each mechanism happens to carry the route most important to its own regime: prompt NO matters in the rich zones where Jet-A burns, and the NNH route matters in the lean premixed conditions where LNG burns. That is fortunate, but it is not a reason to ignore it.

Engineering caution. Because the two fuels use mechanisms with different pathway coverage, *absolute* NO_x values are not strictly comparable between Jet-A and LNG. The design metric defined later compares each fuel only against itself, which avoids this problem entirely.

Rather than demanding a fixed species list, the code records which routes each mechanism carries and attaches that record to every result. A hard requirement for NNH would wrongly reject the Jet-A mechanism, which is otherwise the better of the two. Implemented by `nox_pathway_coverage` in `src/fuelnozzle/crn/chemistry.py`.

5.5 What is checked, and what merely warns

A missing nitrogen submodel *raises an error* rather than issuing a warning. Such a mechanism reports exactly zero NO with no other symptom, so a warning could be missed and the resulting zeros believed. The same applies to a fuel species absent from the mechanism, and to an LNG component that the mechanism cannot represent: dropping a component would silently change the fuel being modeled.

Conditions outside a mechanism's declared validity range produce warnings rather than errors, because extrapolation is sometimes the only option and the user is entitled to make that choice knowingly. GRI-Mech 3.0 was optimized against data up to roughly 10 atm; at combustor pressures of 10 to 40 atm every LNG result carries a warning to that effect. Implemented by `validate_mechanism`.

6 Air-Side Bookkeeping and Equivalence Ratio

6.1 Equivalence ratio

The *equivalence ratio* ϕ says how rich or lean a mixture is. It is the actual fuel-to-air mass ratio divided by the ratio that would consume both exactly:

$$\phi = \frac{(\dot{m}_f/\dot{m}_a)}{(\dot{m}_f/\dot{m}_a)_{\text{stoich}}} = \frac{\dot{m}_f \text{AFR}_{\text{stoich}}}{\dot{m}_a}. \quad (1)$$

$\phi < 1$ is lean, $\phi = 1$ is stoichiometric, and $\phi > 1$ is rich. Peak temperature, and therefore most NO_x, occurs slightly rich of stoichiometric.

Computing ϕ by counting unburned fuel and air fails once reaction has started, since the fuel is gone but the mixture has not changed identity. The code therefore uses an *element-based* definition, which counts carbon, hydrogen, and oxygen atoms rather than molecules and so remains correct in burned, partially burned, and evaporating mixtures alike. The related *Bilger mixture fraction* Z measures the same thing on a scale running from 0 in pure air to 1 in pure fuel, and the two are related through the stoichiometric mixture fraction Z_{st} :

$$\phi = \frac{Z}{1-Z} \cdot \frac{1-Z_{st}}{Z_{st}}, \quad Z_{st} = \frac{1}{1+\text{AFR}_{\text{stoich}}}. \quad (2)$$

Both are evaluated by Cantera’s own routines rather than reimplemented here. Verified against hand calculation: methane gives $\text{AFR}_{\text{stoich}} = 17.13$ and the Jet-A surrogate $\text{C}_{11}\text{H}_{22}$ gives 14.69, and Equation (2) reproduces a prescribed ϕ to six decimal places.

6.2 Where the air goes

Combustor air does not all enter at one place. It is divided among the dome, the primary zone, quench jets, dilution jets, and liner cooling films:

$$f_{\text{dome}} + f_{\text{primary}} + f_{\text{quench}} + f_{\text{dilution}} + f_{\text{cooling}} = 1. \quad (3)$$

The code refuses a split that does not satisfy Equation (3) rather than renormalizing it, because a split that does not sum to one usually means a station was forgotten, and silently rescaling would hide the mistake.

6.3 The inactive circuit, and why it is a design lever

Jet-A and LNG have separate injectors. Air flows through *both* injector air passages at all times, including the one whose fuel is shut off. If the Jet-A passage carries a share a_J of the dome air and the LNG passage carries a_L , with $a_J + a_L = f_{\text{dome}}$, then the air setting the equivalence ratio near the flame depends on which fuel is burning:

$$f_{\text{near}}(\text{fuel}) = a_{\text{active}} + \chi a_{\text{idle}}, \quad (4)$$

where $\chi \in [0, 1]$ is the fraction of the idle passage’s unfueled air that actually reaches the near field.

Equation (4) is the reason a single combustor may be able to run rich for Jet-A and lean for LNG. Making a_J small and a_L large produces a rich Jet-A near field and a lean LNG near field with no moving parts and no change in hole area.

Engineering caution. The strength of this effect depends entirely on χ , and a reactor network cannot determine χ — it is a mixing question that requires computational fluid dynamics or an experiment. Every result that depends on it is reported with a sensitivity across the full range $\chi \in [0, 1]$, never at a single assumed value. At $\chi = 0$ the lever is at full strength; at $\chi = 1$ it disappears entirely.

Implemented by `AirSplit` and `resolve_air_streams` in `src/fuelnozzle/crn/streams.py`.

6.4 Reporting emissions

Raw concentrations are not comparable between combustors, because a leaner combustor dilutes its own exhaust. Two conventional corrections are applied. The *emission index* expresses output per unit of fuel burned,

$$\text{EI}_i = 1000 \frac{Y_i \dot{m}_{\text{exhaust}}}{\dot{m}_{\text{fuel}}} \quad [\text{g/kg}], \quad (5)$$

and concentrations are quoted dry, with water removed, corrected to a reference oxygen level of 15%:

$$\text{ppmv}_{15\%} = 10^6 X_i^{\text{dry}} \frac{20.9 - 15}{20.9 - X_{\text{O}_2}^{\text{dry}} \times 100}. \quad (6)$$

Equation (6) is undefined when the measured oxygen approaches that of air, which happens as combustion approaches extinction; the code raises rather than returning a large meaningless number.

7 Spray Boundary: Connecting the Nozzle to the Network

A spray is described to the network by how much fuel arrives as vapor, and by a set of *droplet classes*: groups of droplets sharing a size, temperature, and velocity. The number of droplets follows from the liquid mass flow and the mass of one droplet,

$$\dot{N}_d = \frac{\dot{m}_\ell}{\rho_\ell \frac{4}{3}\pi r_0^3}. \quad (7)$$

For LNG, part of the fuel has already flashed to vapor inside the nozzle. Only the remainder becomes droplets. Treating all of it as liquid would ask the model to perform atomization work the flash has already done.

7.1 Where the initial size comes from

Three policies are available. The default takes the initial radius as half the Sauter mean diameter from the nozzle model, which uses information the reference paper did not have, since it solves the atomizer separately. The second follows the paper: start at the orifice radius and let aerodynamic breakup find the size. The third is an explicit user-supplied value.

Engineering caution. The existing package refuses to report a Sauter mean diameter without a hardware calibration. That refusal is inherited here. If no calibration exists, the policy silently degrading to a fabricated size would be worse than useless, so instead it falls back to the orifice-radius approach and emits a warning stating that atomization-quality results must not be reported from it.

7.2 Representing a range of sizes

A real spray is not one size. The largest droplets survive far longer than the mean and are the ones that actually reach the flame, so representing only the mean understates how much liquid arrives unevaporated. The Rosin-Rammler distribution describes the mass fraction contained in droplets smaller than D as

$$Q(D) = 1 - \exp\left[-\left(\frac{D}{X}\right)^n\right], \quad (8)$$

where X is a characteristic diameter and n measures how narrow the spray is. Its Sauter mean diameter is

$$D_{32} = \frac{X}{\Gamma(1 - \frac{1}{n})}, \quad (9)$$

which is inverted so that the classes reproduce the Sauter mean diameter requested. Equation (9) diverges as $n \rightarrow 1$, so a lower bound on n is enforced rather than allowing a distribution with no finite Sauter mean.

8 Droplet Breakup

A droplet moving quickly through gas is deformed by the pressure difference across it and pulled back into shape by surface tension. The Taylor analogy treats that competition as a mass on a spring with friction. Writing y for the distortion scaled so that $y = 1$ means the deformation has reached the droplet radius,

$$\frac{d^2y}{dt^2} = \frac{C_f}{C_b} \frac{\rho_g u_d^2}{\rho_\ell r_0^2} - \frac{C_k \sigma}{\rho_\ell r_0^3} y - \frac{C_d \mu_\ell}{\rho_\ell r_0^2} \frac{dy}{dt}, \quad (10)$$

with $C_k = 8$, $C_f = 1/3$, $C_d = 5$, $C_b = 1/2$. The three terms are, in order, the aerodynamic forcing, the surface-tension restoring force, and viscous damping. The balance between forcing and surface tension is the Weber number,

$$\text{We}_g = \frac{\rho_g u_d^2 r_0}{\sigma}, \quad (11)$$

and the droplet breaks when the distortion reaches unity. Under steady forcing the distortion oscillates about an equilibrium value $\text{We}_c = \text{We}_g/12$ with amplitude A , so breakup is possible when $\text{We}_c + A > 1$.

Engineering caution. John et al. print this criterion as $A + \text{We}_g > 1$. That cannot be what they computed: for their own Fig. 3 case $\text{We}_g \approx 38$, so the criterion would fire immediately and no breakup time would be resolvable, yet their figure shows a finite 2×10^{-5} s. The standard form $\text{We}_c + A > 1$ is used here and reproduces a breakup time of the right order.

The child radius follows from an energy balance between the parent's distortion energy and the new surface created,

$$r = \frac{r_0}{1 + \frac{8K}{20} y^2 + \frac{\rho_\ell r_0^3 y^2}{\sigma} \frac{6K - 5}{120}}, \quad K = \frac{10}{3}. \quad (12)$$

One pass rarely reaches a stable size, so each child becomes the new parent and the test repeats until the Weber number is too small to sustain further breakup.

Engineering caution. The Taylor analogy describes a droplet torn apart from outside by aerodynamic forces. A flashing LNG droplet is burst from within by vapor generation, which is a different mechanism entirely. Breakup is therefore not applied in the flashing regimes; the size comes from the Tier 3 flash-spray result instead, and the decision is recorded as a warning rather than left implicit.

9 Droplet Evaporation

Evaporation can be limited by either of two different things, and which one applies decides the equations.

9.1 Diffusion limited

Below the boiling point, fuel vapor must diffuse away from the surface before more can leave. The surface vapor mass fraction follows from the vapor pressure,

$$Y_s = \frac{W_F}{W_F + W_{\text{mix}} \left(\frac{p_g}{p_v(T_d)} - 1 \right)}, \quad (13)$$

and the difference between surface and far-field concentration drives the transfer,

$$B_d = \frac{Y_s - Y_\infty}{1 - Y_s}, \quad \text{Sh} = \left(2 + 0.6 \text{Re}_d^{1/2} \text{Sc}^{1/3} \right) \frac{\ln(1 + B_d)}{B_d}, \quad (14)$$

$$\frac{dr}{dt} = -\frac{\rho_g D}{2\rho_\ell r} B_d \text{Sh}. \quad (15)$$

In still gas this reduces to $d(r^2)/dt = \text{constant}$, the classical D^2 law: the square of the diameter falls linearly with time. Any evaporation model must reproduce it, and this one is verified to do so — but only after the droplet has finished warming up, since the film temperature moves during that transient.

9.2 Boiling limited

Equation (13) gives $Y_s = 1$ when the vapor pressure reaches the ambient pressure, at which point B_d becomes infinite and the diffusion description fails. That is the boiling point, and it is exactly where the second branch takes over.

A boiling droplet cannot get any hotter: its temperature is pinned at saturation and every joule that arrives goes into the phase change,

$$\frac{dm_d}{dt} = -\frac{\dot{Q}_{\text{conv}}}{L_{\text{vap}}}. \quad (16)$$

Engineering caution. This branch is the single most important addition to the reference model. Flashing LNG arrives above its saturation temperature, so it is *never* diffusion limited. Applying Equation (15) to it would be wrong at the first step.

10 Droplet Heating and Energy Accounting

Heat reaches the droplet by convection,

$$\dot{Q}_{\text{conv}} = \frac{A_d \beta_{\text{spray}} \text{Nu}_d k(T_g - T_d)}{2r}, \quad \text{Nu}_d = \left(2 + 0.6 \text{Re}_d^{1/2} \text{Pr}_d^{1/3} \right) \frac{\ln(1 + B)}{B}, \quad (17)$$

with gas properties evaluated at the film temperature $\bar{T} = (T_g + 2T_d)/3$, weighted toward the droplet because most of the resistance sits in the thin layer against the surface.

Part of that heat drives the phase change and the rest warms the liquid:

$$c_\ell m_d \frac{dT_d}{dt} = \dot{Q}_{\text{conv}} + \frac{dm_d}{dt} L_{\text{vap}}. \quad (18)$$

Since dm_d/dt is negative, the second term subtracts, as it must.

10.1 The bookkeeping that is easy to get wrong

Three statements must hold together, and stating them separately is what keeps the energy balance honest:

- (i) the gas *loses* $\dot{Q}_{\text{conv}}$;
- (ii) the droplet *gains* $\dot{Q}_{\text{conv}}$, spends $|dm_d/dt|L_{\text{vap}}$ on vaporization, and warms with the remainder;
- (iii) the vapor enters the gas carrying the enthalpy it has *at the droplet temperature* T_d , not at the gas temperature.

If item (iii) is neglected and the vapor is added at gas temperature, the latent heat is counted twice and the reactor runs hot. Equation (18) is verified directly: the convective heat equals the sum of the latent and sensible terms to ten significant figures.

11 Reactor Types and the Cantera Implementation

A *reactor* here is a region of the combustor treated as internally uniform. Which idealization it is given states what is being assumed about mixing inside it. A *perfectly stirred reactor* assumes anything entering is instantly spread throughout, which suits recirculation and flame zones where turbulence is violent. A *plug flow reactor* assumes the flow marches along without mixing backwards, which suits the post-flame length where axial velocity dominates. An *evaporator* and a *mixer* are stirred reactors that additionally contain droplets.

Plug flow is represented as a chain of stirred reactors rather than by Cantera's `FlowReactor`, so that it participates in the same simultaneous network solution as everything else. The number of links is a numerical parameter whose convergence must be demonstrated, not assumed.

The residence time of a zone is

$$\tau_i = \frac{\bar{\rho}_i V_i}{\dot{m}_i}. \quad (19)$$

Engineering caution. Because a reactor stands for a coarse region with real internal variation, Equation (19) gives an *effective zone-scale* residence time. It is not the transit time of any particular parcel of fluid, and should not be read as one.

12 Network Assembly, Mass Balance, and Steady Solution

12.1 The splits will not balance, and what to do about it

Flow splits supplied by a user, or extracted from a computational fluid dynamics solution, will not conserve mass exactly. John et al. report imbalances up to 9.3% on a single reactor. For each zone, what enters must equal what leaves:

$$\dot{m}_{i,\text{in}} + \sum_j \dot{m}_{ji} = \dot{m}_{i,\text{out}} + \sum_j \dot{m}_{ij}. \quad (20)$$

Collecting the reactor-to-reactor flows into a vector $\mathbf{z}$, that is a linear system $\mathbf{A}\mathbf{z} = \mathbf{b}$, which the supplied flows miss by a residual $\mathbf{r} = \mathbf{b} - \mathbf{A}\mathbf{z}$. We want the *smallest* change $\Delta\mathbf{z}$ that repairs it:

$$\Delta\mathbf{z}^* = \arg \min \|\Delta\mathbf{z}\|_2^2 \quad \text{subject to} \quad \mathbf{A} \Delta\mathbf{z} = \mathbf{r}. \quad (21)$$

There are normally more flows than reactors, so the system is underdetermined and Equation (21) is exactly what a least-squares solve returns. Choosing the smallest change rather than rescaling means no single flow absorbs the entire error, and flows that were already consistent are barely touched.

The rows of **A** sum to zero, since every internal flow leaves one reactor and enters another. A solution therefore exists only if the external streams balance globally. When they do not, the boundary conditions themselves are wrong and no redistribution can help, so the code raises rather than returning a plausible-looking repair.

12.2 Solving the network

Recirculation is what makes this a graph rather than a chain: hot products flowing back toward the injector are what hold the flame, so a zone's inlet can depend on its own downstream. All zones are therefore handed to a single Cantera **ReactorNet** and solved together, which handles the recycle without any iteration on our part.

The steady state is reached by *marching in time* until nothing moves any more, rather than by calling Cantera's steady-state solver.

Engineering caution. This reverses the original plan, for a measured reason. On a four-reactor, 71-species recirculating network, `advance_to_steady_state` ran for 210 s and then failed. The same case marches to a converged answer, with an identical peak temperature of 1881 K, in 0.86 s. Marching is also bounded and observable: it integrates in chunks of two residence times, stops when every reactor temperature moves less than 0.05 K, and gives up after two hundred residence times with a warning. The cost is that element conservation closes to about 10^{-7} rather than 10^{-11} , reflecting that tolerance.

12.3 Detecting a solution that is not a combustor

A network initialized cold converges to a perfectly valid steady state in which nothing is burning. It solves the equations and it is useless. Reactors are therefore started hot and equilibrated, and any converged state whose peak temperature is within 50 K of the inlet air is reported as extinguished, because emissions computed from it are meaningless.

13 Coupling the Droplets to the Network

Evaporation depends on gas temperature, and gas temperature depends on evaporation. Solving both at once would place a stiff droplet system inside a stiff chemistry system. Instead the two are alternated: the droplets are marched with the gas held fixed, the network is solved with the resulting sources held fixed, and the process repeats with only part of each update applied.

The partial step matters. Evaporation and temperature reinforce each other, since hotter gas evaporates more fuel, which burns and makes the gas hotter still, so taking the full step invites oscillation between a flooded and a starved state.

Two details keep this honest. The vapor enters each zone at the *droplet* temperature, and the heat the droplets absorbed is removed from the gas; without the second, injecting vapor would hand the gas back the latent heat it had just spent. And the network is rebuilt rather than edited on each pass, because evaporating fuel changes the total gas mass flow and therefore the mass balance itself.

Droplets surviving the whole spray path are reported as *liquid carryover*. This is a design failure rather than a rounding detail: fuel that never evaporates never burns.

14 Combustor Architecture Templates

The architectures differ in one decision, how much air meets the fuel before it burns, and that decision is what the dual-fuel study exists to make.

14.1 Rich-quench-lean

Burn the fuel rich, where it is too fuel-heavy to get very hot, then add air fast enough to cross stoichiometric quickly and finish lean.

Engineering caution. The quench is modeled as a chain of stages, not a single mixing point. This is not cosmetic. Nitric oxide is produced *while* the mixture crosses stoichiometric on its way from rich to lean, and a single perfectly mixed point would jump straight over that crossing and hide the dominant source entirely.

Quench air is distributed along the stages by a declared schedule. Front-loading shortens the traverse but risks quenching the reaction; rear-loading does the opposite. The sensitivity to stage count and schedule is a required output, not an option.

14.2 Lean premixed and lean direct injection

Lean premixed mixes fuel and air completely before burning, so no part of the flame is hot enough to make much nitric oxide. Lean direct injection sprays into a lean dome with no premixing passage at all.

Engineering caution. Chemistry is *not* frozen in the premixer, even though the whole idea of a premixer is that nothing reacts inside it. Freezing it would guarantee the assumption instead of testing it. Left reacting, the premixer ignites in the model precisely when residence time exceeds ignition delay, which is the real failure mode of the architecture. Lean direct injection has no premixing passage and therefore nowhere for this to happen, which is exactly why it is the natural comparison for Jet-A at landing and take-off.

14.3 The idle passage as a design lever

Because the two fuel circuits are separate hardware, air flows through *both* injector passages at all times, including the one whose fuel is shut off. If the Jet-A passage carries a share a_J of the dome air and the LNG passage a_L , the air setting the equivalence ratio near the flame is

$$f_{\text{near}} = a_{\text{active}} + \chi a_{\text{idle}}, \quad (22)$$

where χ is the fraction of idle-passage air reaching the near field. Every template routes this explicitly. Making a_J small and a_L large gives a rich Jet-A near field and a lean LNG near field with no moving parts.

Engineering caution. At $\chi = 0$ the lever is at full strength; at $\chi = 1$ it disappears entirely. A reactor network cannot determine χ , so every result depending on it is reported across its full range rather than at one assumed value.

14.4 Merging mechanisms

Combining a fuel mechanism with a nitrogen submodel is sometimes necessary, and the tool checks such a combination without performing it, because the choice changes the answer and belongs to whoever is accountable for it. Checked against the two mechanisms actually in use, it finds six species with inconsistent thermodynamic data, 223 duplicate reactions, and a species spelled CH2* in one file and CH2(S) in the other. The last is an error rather than a warning: a merged mechanism would carry two separate pools of what is physically one species.

15 Autoignition and Flashback Screening

A premixing passage exists so that fuel and air are uniform before they burn. It works only if the mixture takes longer to ignite on its own than it takes to cross the passage. The margin is a ratio of two times,

$$M = \frac{\tau_{\text{ign}}}{\tau_{\text{res}}}, \quad (23)$$

and a design guidance floor of $M \geq 4$ is used as an editable input rather than a hidden constant. When $M < 1$ the mixture ignites inside the injector, and the result is reported as an error rather than a number.

15.1 Why cold fuel buys margin

Fuel that has flashed arrives cold, and whatever is still liquid draws its latent heat out of the air. Both effects lower the premixed temperature, and because ignition delay depends on temperature exponentially, a modest cooling buys a large increase in margin. Measured here with air at 800 K and 20 atm:

Case	T_{mix}	Drop	of which latent
Jet-A vapour at 470 K	774.1 K	25.9 K	0.0 K
Jet-A, 30% liquid	771.5 K	28.5 K	2.6 K
LNG vapour at 150 K	747.2 K	52.8 K	0.0 K
LNG, 30% liquid	742.8 K	57.2 K	4.4 K

LNG cools the mixture about twice as much as Jet-A. Note that the latent contribution is only about 4 K of a 57 K drop: the benefit is overwhelmingly the sensible enthalpy of cold vapour, not the phase change. It would be easy, and wrong, to assume the opposite.

15.2 What that is worth

Fuel	T_{mix}	τ_{ign}	M at 1 ms	M at 3 ms
Jet-A	774 K	4.04 ms	4.0 (safe)	1.3 (marginal)
LNG	747 K	no ignition in 10 s	unbounded	unbounded

Jet-A tolerates roughly one millisecond of premixing at landing and take-off conditions. LNG is effectively inert on premixer timescales. This is the quantitative basis for considering a lean premixed LNG cruise alongside a lean direct injection or rich-quench-lean Jet-A landing and take-off.

Engineering caution. The flashback screen is *suppressed* unless a laminar flame speed is supplied. Obtaining one requires a flame calculation this tool does not perform, and inventing a value would place an unmarked guess underneath a safety screen. The screen reports nothing rather than something unfounded.

16 Cryogenic Thermal Fuel Management

LNG leaves the tank near 110 K and must reach the injector warm enough to flash usefully. Fuel temperature is therefore a design variable, pulled in four directions at once: enough superheat at the nozzle for the intended flash regime; enough subcooling upstream that the feed line does not boil; enough autoignition margin; and within the heat actually available to dump into the fuel.

16.1 The window, and what sets its width

Superheat at the injector and subcooling in the line compete for the same resource: the temperature gap between saturation at feed pressure and saturation at chamber pressure. With a 20 bar chamber:

Feed pressure	Saturation gap	Feasible T_{fuel}	Width
2.5 MPa	6.1 K	none	—
3.0 MPa	11.4 K	171.0–172.0 K	1.0 K
4.0 MPa	20.2 K	171.0–181.0 K	10.0 K

Engineering caution. Below roughly 3 MPa feed pressure there is *no* feasible fuel temperature at all. No amount of heating fixes this, because the gap itself is too narrow. Pump pressure is what buys thermal design freedom, which connects this window directly to the pressure budget the package already computes.

Above the critical pressure of the mixture there is no saturation at all. Supercritical injection is a legitimate design choice, but the subcooling and superheat constraints are defined only below the critical point and do not apply to it; the code says so explicitly rather than failing obscurely.

16.2 The circuit that is not running

A dual-fuel nozzle always has one circuit hot and stagnant. With no flow there is no convective cooling, so the idle fuel sits near wall temperature. Jet-A left in a hot passage forms deposits that block the injector; LNG left in one boils, and restart then delivers vapour rather than liquid. Either can veto an otherwise attractive packaging, and it is cheaper to discover here than on a test stand.

17 Verification Coverage

Verification asks whether the software solves the equations it claims to. It is kept separate from validation, which asks whether those equations describe reality.

17.1 Limits and identities

The residence-time definition reproduces the identity the reference paper states. A stirred reactor given a long residence time approaches the equilibrium state. Shrinking it below a critical size extinguishes it. With no fuel, air passes through unchanged. Droplet evaporation recovers the D^2 law once the warm-up transient is excluded, and the boiling branch reproduces $\dot{m} = \dot{Q}/L$ exactly. The droplet energy split closes to ten significant figures. Element balances close to about 10^{-7} .

17.2 Convergence

A plug flow zone represented as a chain of stirred reactors converges with the number of links. The droplet integration is insensitive to solver tolerance across four orders of magnitude. The ignition-delay table changes by under 15% when its temperature grid is refined fourfold.

17.3 Comparison with the reference paper

The purpose is directional: to confirm the implementation reproduces the paper’s trends and conclusions, not its absolute numbers. Two of its quantitative anchors are only partly reproducible, and this is stated rather than smoothed over.

Breakup and evaporation timescales. Breakup agrees to an order of magnitude, at 3.4×10^{-5} s against their 2×10^{-5} s. Evaporation does not: we obtain 0.80 s against their reported 6 ms at 350 K. Their case is underspecified, giving neither ambient pressure nor liquid properties, and 6 ms is difficult to reconcile with realistic kerosene volatility. What is reproduced is their *conclusion*: breakup is faster than evaporation by more than two orders of magnitude at 350 K, the separation narrows with temperature, and it still exceeds one order of magnitude at 900 K. That conclusion is what justifies treating the two sequentially, which is what the implementation depends on.

Liquid against gaseous networks. This is reproduced cleanly. The same topology and flows, run once with droplets and once with the fuel gaseous and perfectly mixed:

	ϕ range	Peak T	Exit T	NO
With spray	0.514–1.714	2453 K	1926.6 K	1868 ppm
Gaseous, premixed	0.514–0.514	1928 K	1928.0 K	57 ppm

Exit temperature differs by 0.07%. Nitric oxide differs by a factor of 33. That asymmetry is the paper’s entire argument: both methods get the global energy balance right, and only one gets the emissions right, because nitric oxide responds exponentially to the hot pockets that spray heterogeneity creates. Their reported contrast, a gaseous network underpredicting by 54–91% with peak temperatures near 1800 K against above 2200 K with spray, matches this in both direction and character.

Engineering caution. Achieving this required the gaseous comparison to be genuinely premixed, with fuel distributed to every zone in proportion to its air. Injecting the vapour only at the dome retains the air-staging heterogeneity and is not the paper’s comparison; the two cases then differ by a few percent instead of a factor of thirty.

17.4 What is not claimed

Absolute nitric oxide is not claimed to match the paper. Their reactor volumes, connectivity, spray path lengths, and propagation fractions are unpublished, so their calibrated network cannot

be rebuilt. Their chemistry also differs: they used a mechanism whose oxidation base carries no nitrogen chemistry, and the paper does not state which nitrogen submodel they appended.

Absolute emission levels from this implementation are also not validated, and the reason is now understood rather than merely suspected.

17.5 How fast the quench is decides the answer

The rich-quench-lean optimum initially fell at $\phi_{\text{rich}} = 1.22$, richer than classical practice indicates, on a curve so flat it barely had a minimum. A sensitivity study established that the cause was an unrealistically long quench, not the reactor model.

Discretization first. Splitting the quench into stages matters, and the size of the effect is worth stating plainly: a *single* perfectly mixed quench point underpredicts nitric oxide by more than half, because it jumps straight over the stoichiometric crossing where most of it is made. The result converges to about 1% by twelve stages.

Quench *speed* matters more:

Quench residence time	Optimum	Minimum EI _{NOx}
~0.9 ms (realistic)	$\phi_{\text{rich}} = 1.35$, sharp	68.5
~4.6 ms	$\phi_{\text{rich}} = 1.22$, flat	130.6
~18 ms	no interior minimum	176.9

Real quench jets penetrate and mix in roughly half a millisecond to a millisecond, and that speed is the entire point of the architecture. At a realistic quench the optimum lands at 1.35 with a pronounced minimum, meeting the sanity anchor. A slow quench dwells near stoichiometric, inflates nitric oxide, and progressively destroys the very optimum the architecture exists to exploit.

Engineering caution. The optimum itself exists only when there is enough quench *air*, not merely enough quench speed. Measured at four quench-air fractions, an interior optimum is present at 0.30 and 0.38 of the total air but absent at 0.15 and 0.20, where the curve inverts and $\phi_{\text{rich}} \approx 1.4$ becomes the worst choice rather than the best. Rich-quench-lean depends on crossing stoichiometric quickly, and the crossing needs both time and air. Any statement about a rich-zone optimum must name the quench-air fraction it assumes.

Engineering caution. Absolute emission indices remain roughly twice what real rich-quench-lean hardware achieves. The zones here are adiabatic, and a perfectly stirred quench still mixes more slowly than real jets even at one millisecond. The optimum's *location* and the *trends* are supported; the absolute *level* is not.

18 Known Deviations and Technical Debt

The numbered deviations table lives in `docs/CRN_IMPLEMENTATION_LOG.md` and is the authoritative record. The ones that change how a result should be read are:

- Jet-A uses two mechanisms, one for the network and one for ignition delay, because a high-temperature-only mechanism overpredicts ignition delay by $510\times$ at 700 K and would declare an unsafe premixer safe.
- The premixer is modelled as reacting rather than chemically frozen, so that its real failure mode is visible in the solution.

- The breakup criterion follows the standard Taylor analogy rather than the form printed in the paper, which cannot be what they computed.
- Steady state is reached by marching in time rather than by Cantera’s steady solver, which failed on a recirculating network. Element conservation closes to about 10^{-7} rather than 10^{-11} as a result.
- The flashback screen and all droplet-size outputs are suppressed rather than estimated when the inputs they need are absent.

A Cantera Interface Used by This Work

Pinned to **Cantera 3.2.0**. Verified present in the installed version rather than taken from documentation, because the reactor and flow-device interface changed between Cantera 2.6 and 3.x.

Purpose	Cantera name	Note
Gas mixture and kinetics	<code>Solution</code>	One per reactor; never shared
Fixed boundary condition	<code>Reservoir</code>	Air inlets, fuel vapor, exhaust
Perfectly stirred reactor	<code>IdealGasConstPressureMoleReactor</code>	Mole-based; see below
Alternative reactor forms	<code>IdealGasReactor</code> , <code>IdealGasMoleReactor</code>	
Custom governing equations	<code>ExtensibleIdealGasConstPressureReactor</code>	For coupled droplets
Steady plug flow	<code>FlowReactor</code>	Not <code>PlugFlowReactor</code> ; see below
Prescribed stream	<code>MassFlowController</code>	Reactor-to-reactor flows
Pressure closure	<code>PressureController</code>	Network exit
Pressure-driven flow	<code>Valve</code>	
Heat transfer	<code>Wall</code>	Liner heat loss
Surface chemistry	<code>ReactorSurface</code>	Not currently used
Network assembly and solution	<code>ReactorNet</code>	
Steady solution	<code>ReactorNet.advance_to_steady_state</code>	
Time integration	<code>ReactorNet.advance</code> , <code>.step</code>	
Large-mechanism speedup	<code>AdaptivePreconditioner</code>	Mole reactors only
Stoichiometry	<code>Solution.stoich_air_fuel_ratio</code>	
Equivalence ratio	<code>Solution.equivalence_ratio</code>	Element-based
Mixture fraction	<code>Solution.mixture_fraction</code>	Bilger
Profile storage	<code>SolutionArray</code>	

Two findings from that verification are worth recording.

Mole-based reactors are not a stylistic preference. Sparse Jacobian preconditioning, which is what makes a network of 70-species reactors solve in seconds rather than minutes, is supported only for `IdealGasMoleReactor` and `IdealGasConstPressureMoleReactor`. That determines the default reactor type.

There is no `PlugFlowReactor` class. The Cantera documentation describes a “Plug Flow Reactor”, but the class is named `FlowReactor`. A plug flow zone is represented here by a series of stirred reactors instead, which participates in the simultaneous network solution; `FlowReactor` is reserved as an independent cross-check.

B Building This Report

From the repository root:

```
pixi run doc-crn
```

which runs Tectonic on this file. `pixirundoc` builds both this report and the nozzle technical reference.